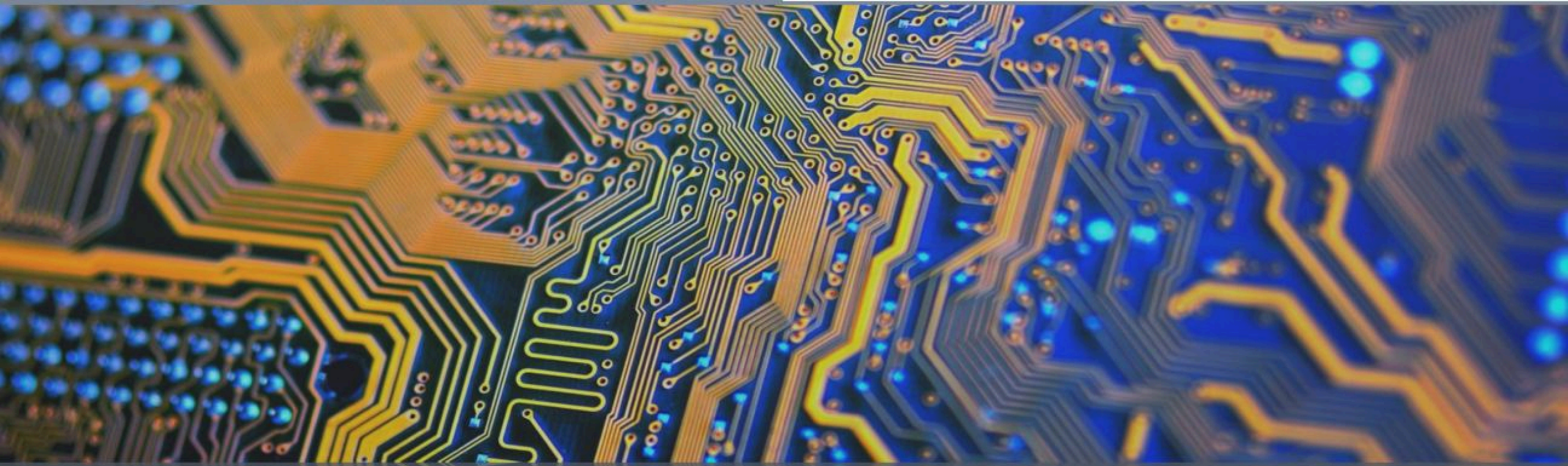


# CEIS101 IOT
# COURSE PROJECT PRESENTATION

# SMART HOME AND AUTOMATION SYSTEM

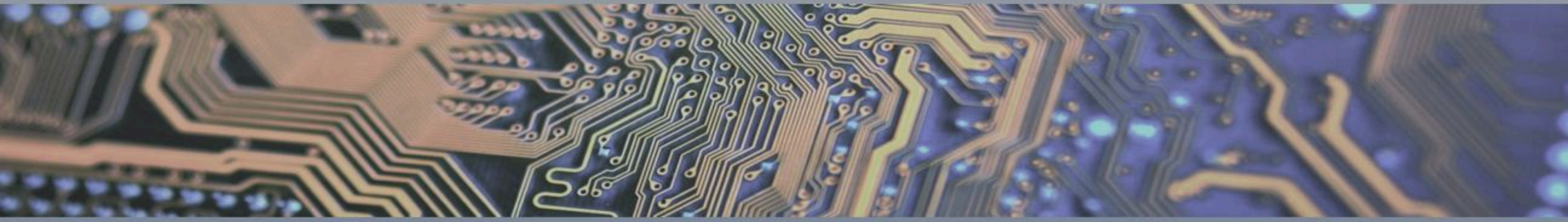
Presented By Mildred Owens

# INTRODUCTION I

## This IoT Course Project in Home Automation System

- COVERS THE FUNDAMENTAL CONCEPT OF THE IOT BY INTEGRATING HARDWARE, SOFTWARE, AND NETWORKS INTO A WHOLE SYSTEM.
- INTRODUCED ME TO THE ELEMENTS OF THE INTERNET OF THINGS (IOT), AUTOMATION, BASIC INTRO TO C++ PROGRAMMING FOR ARDUINO, DESIGNING VIA TINKER CAD, BASIC ELECTRONICS AND TO NETWORK SECURITY.
- PROVIDED ME HANDS-ON EXPERIENCE ON BUILDING MY OWN SMART HOME AUTOMATION AND SECURITY SYSTEM PROTOTYPE WITH THE IOT KIT COMPONENTS BY:

  - simulating and building a circuit in Tinkercad;
  - adding a door sensor to check whether the door is open or closed;
  - adding a distance sensor to monitor for a possible home intruder; and
  - adding an automated light to the smart home.



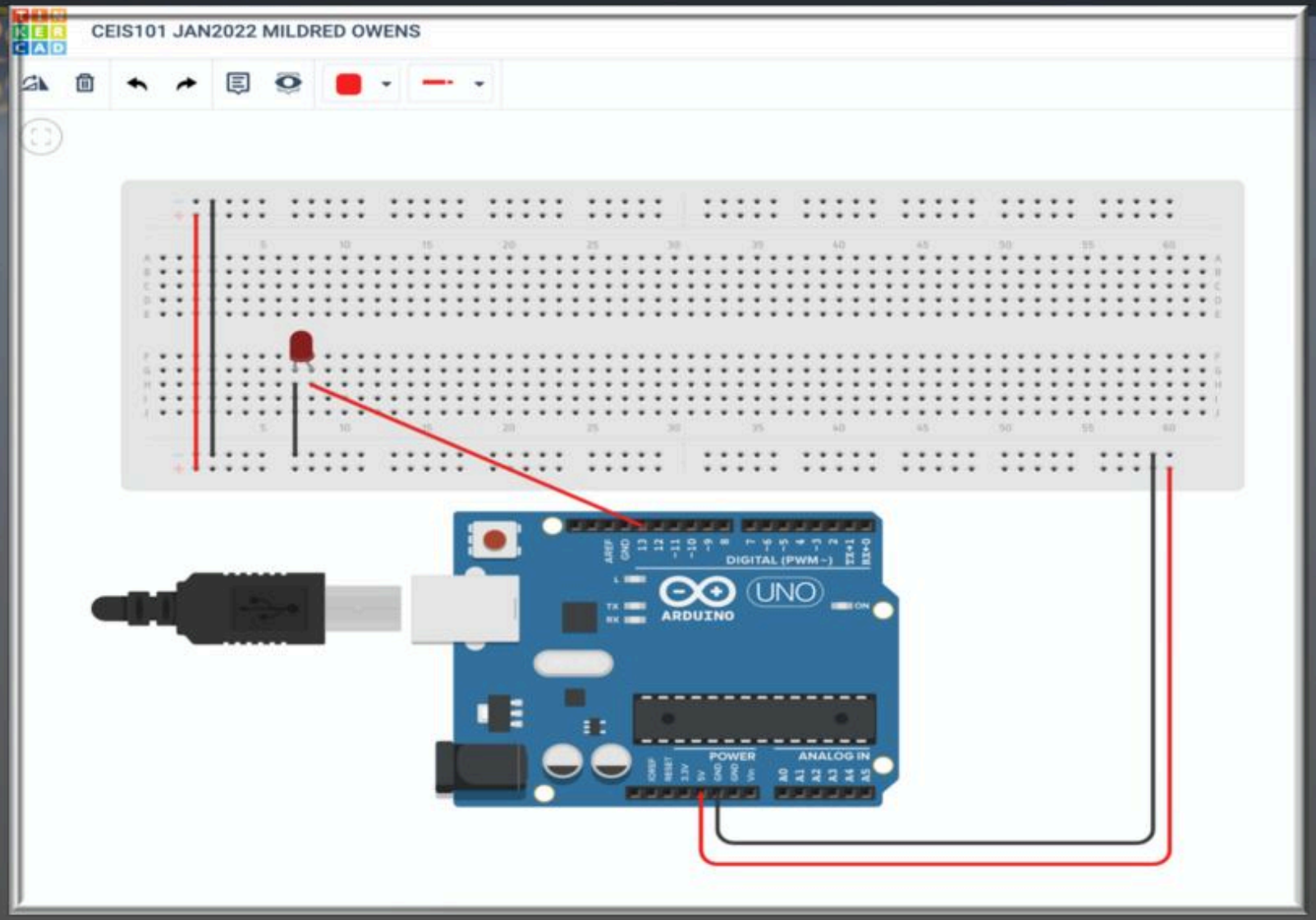
# MODULE 2 : TINKERCAD

## CIRCUIT SIMULATION IN TINKERCAD

# CIRCUIT SCREENSHOT

**Tinkercad helped me**

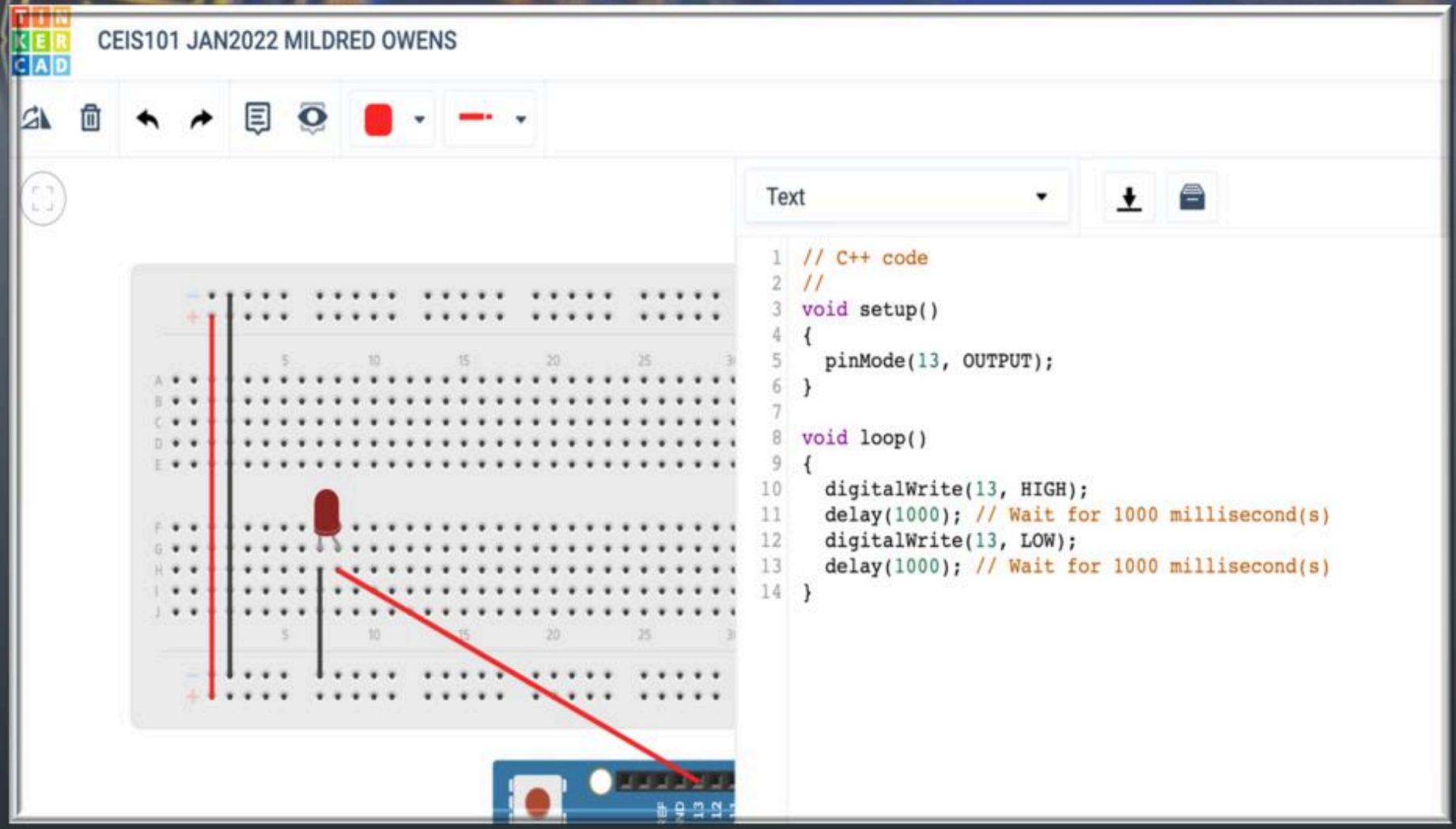
- plan and design the placement of my components, connect and design positions for my interactive circuits (jump wires) to electronic parts from breadboard to Arduino mega2560

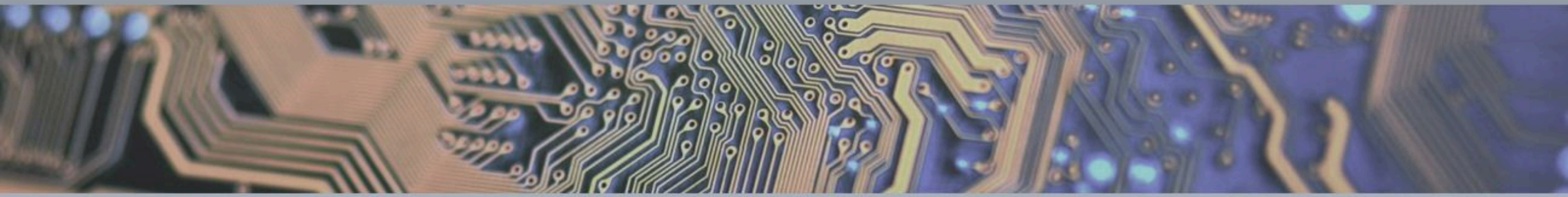


# CODE SCREENSHOT (ARDUINO IDE)

**Connected to the right port via the Computer using Arduino IDE,**

- see a code view, where can I directly edit using either a built-in blocks-style interface, a text-based editor, or a combined view
- test my code and watch for expected results and evaluate/adjust my connections as needed.





# MODULE 3
# INVENTORY/LED CIRCUIT

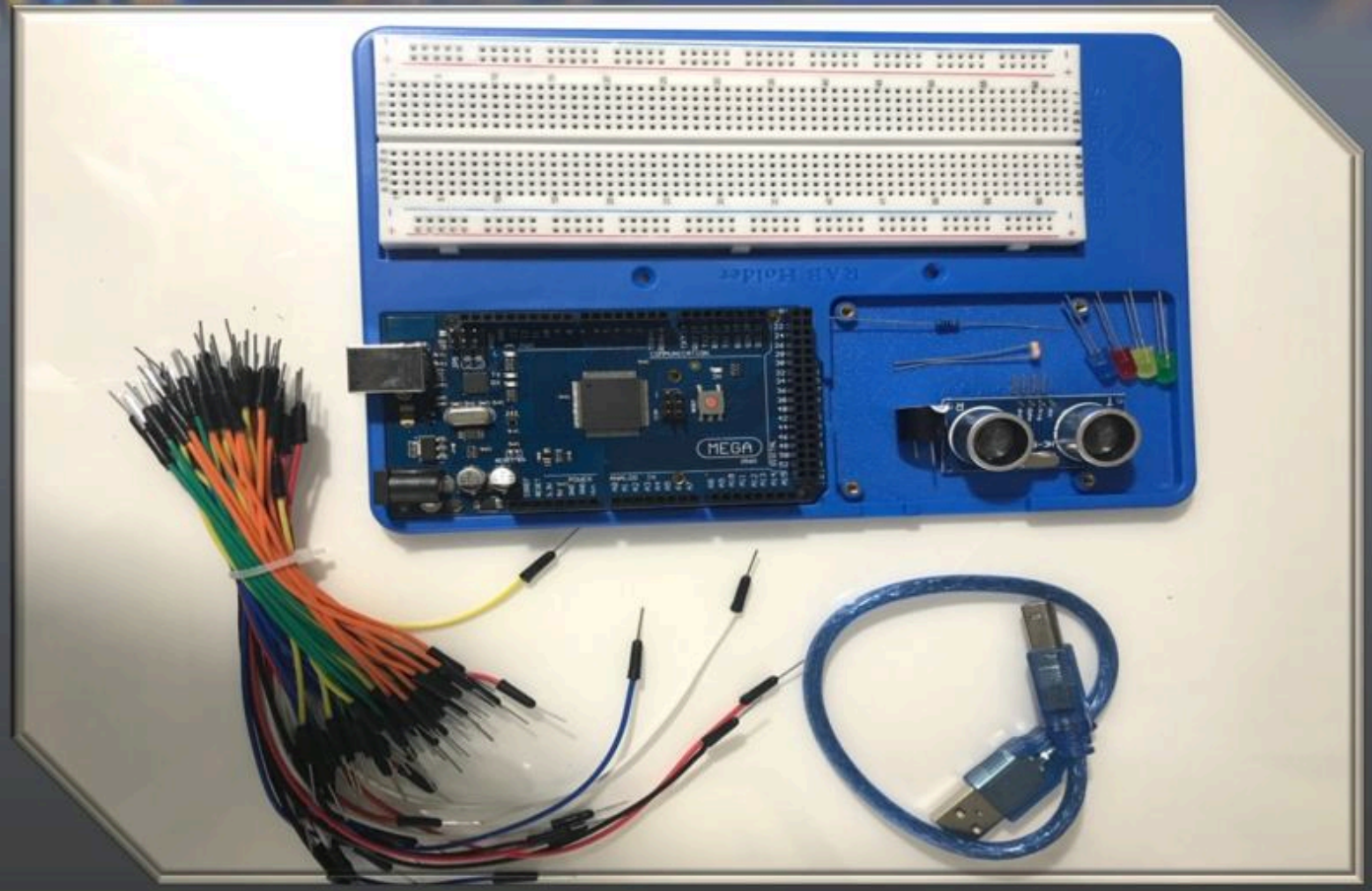
# INVENTORY OF IOT KIT

- UCTRONICS KIT
- ESP32 (2)
- LCD MODULES (2)
- BREADBOARDS (3)
- MINI ROUTER
- PATCH CABLE
- DIGITAL MULTI METER
- USB TO MICRO USB (2)

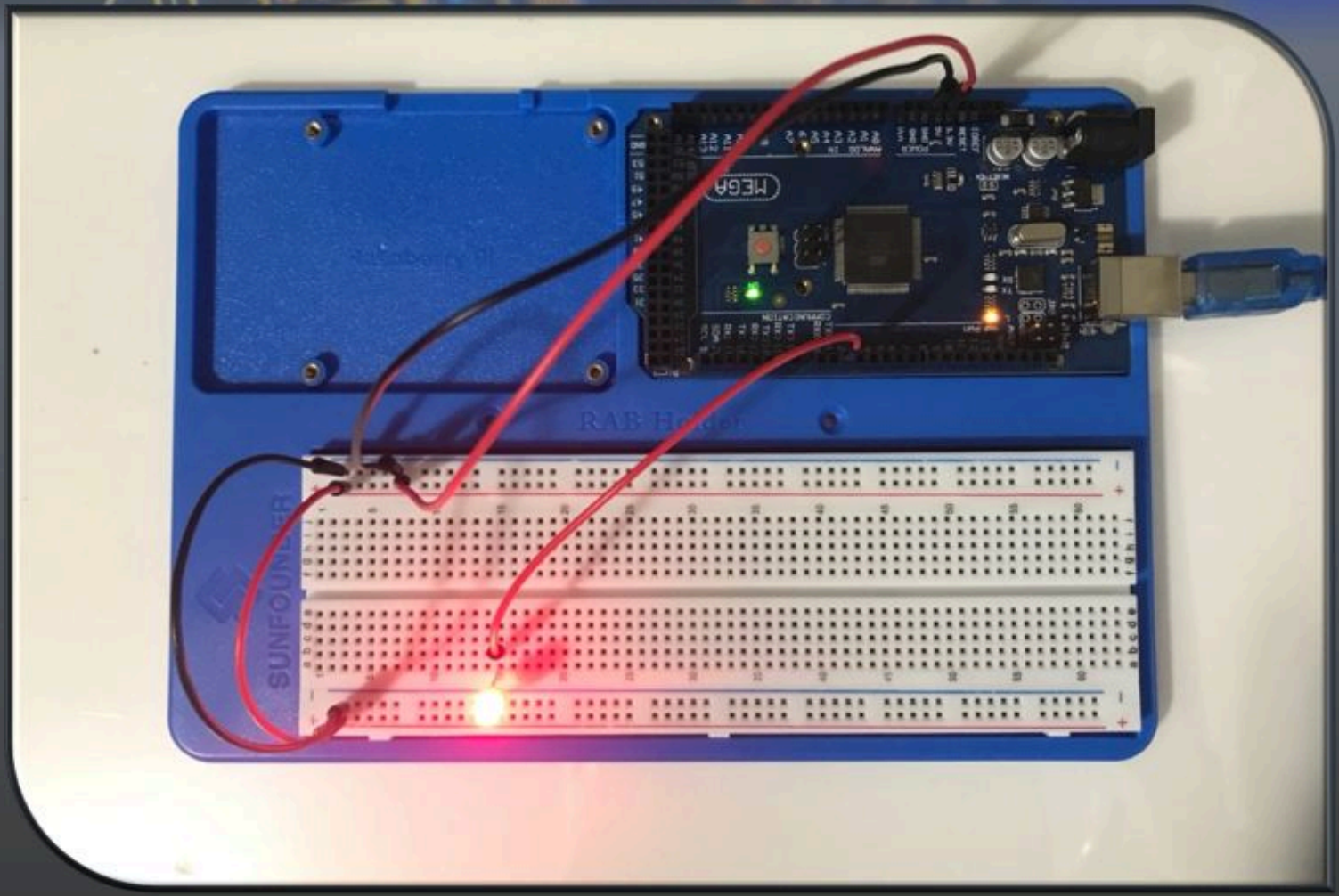


# ORGANIZATION OF PROJECT COMPONENTS

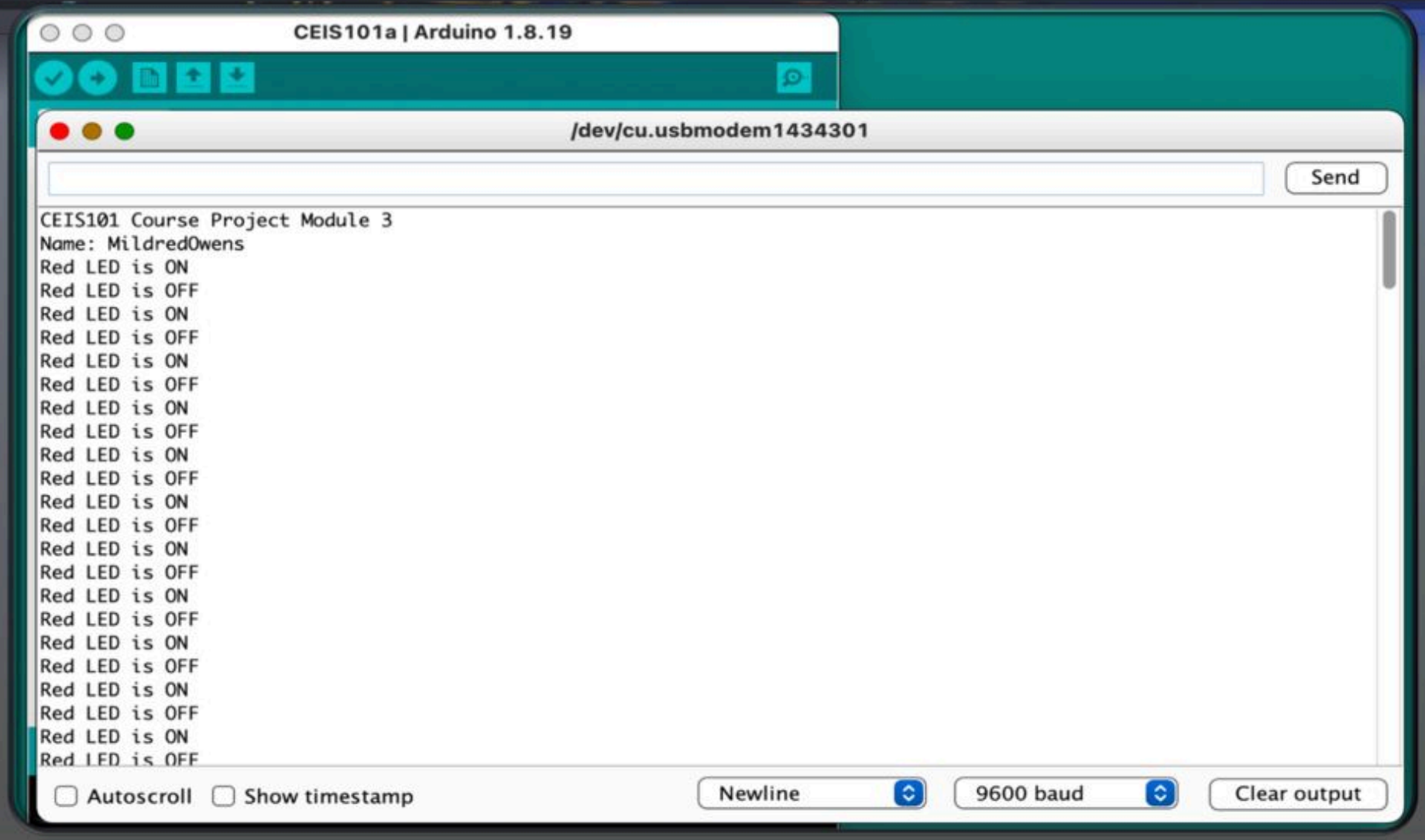
- Arduino Mega 2560
- Breadboard
- Resistor 10kΩ
- LEDs
- Ultrasonic Sensor
- Active Buzzer
- Photoresistor
- Wires
- USB Type B cable

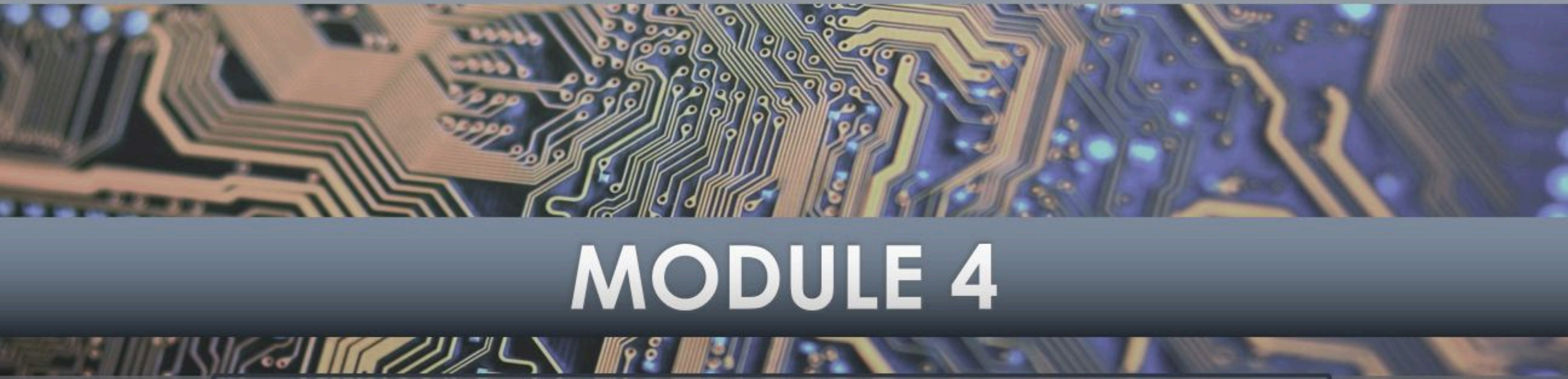


# CIRCUIT WITH RED LED ON



# SERIAL MONITOR (SCREENSHOT)

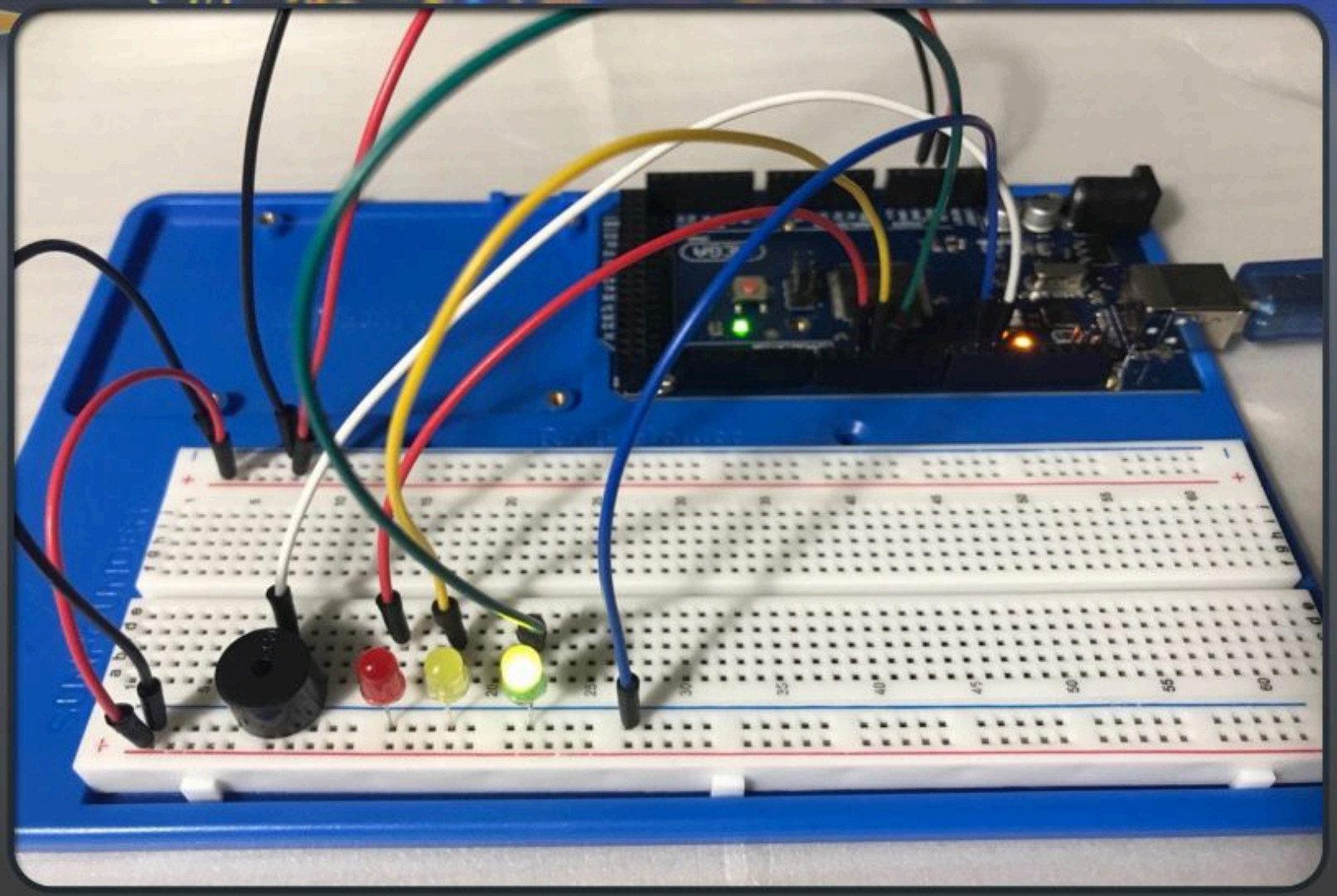




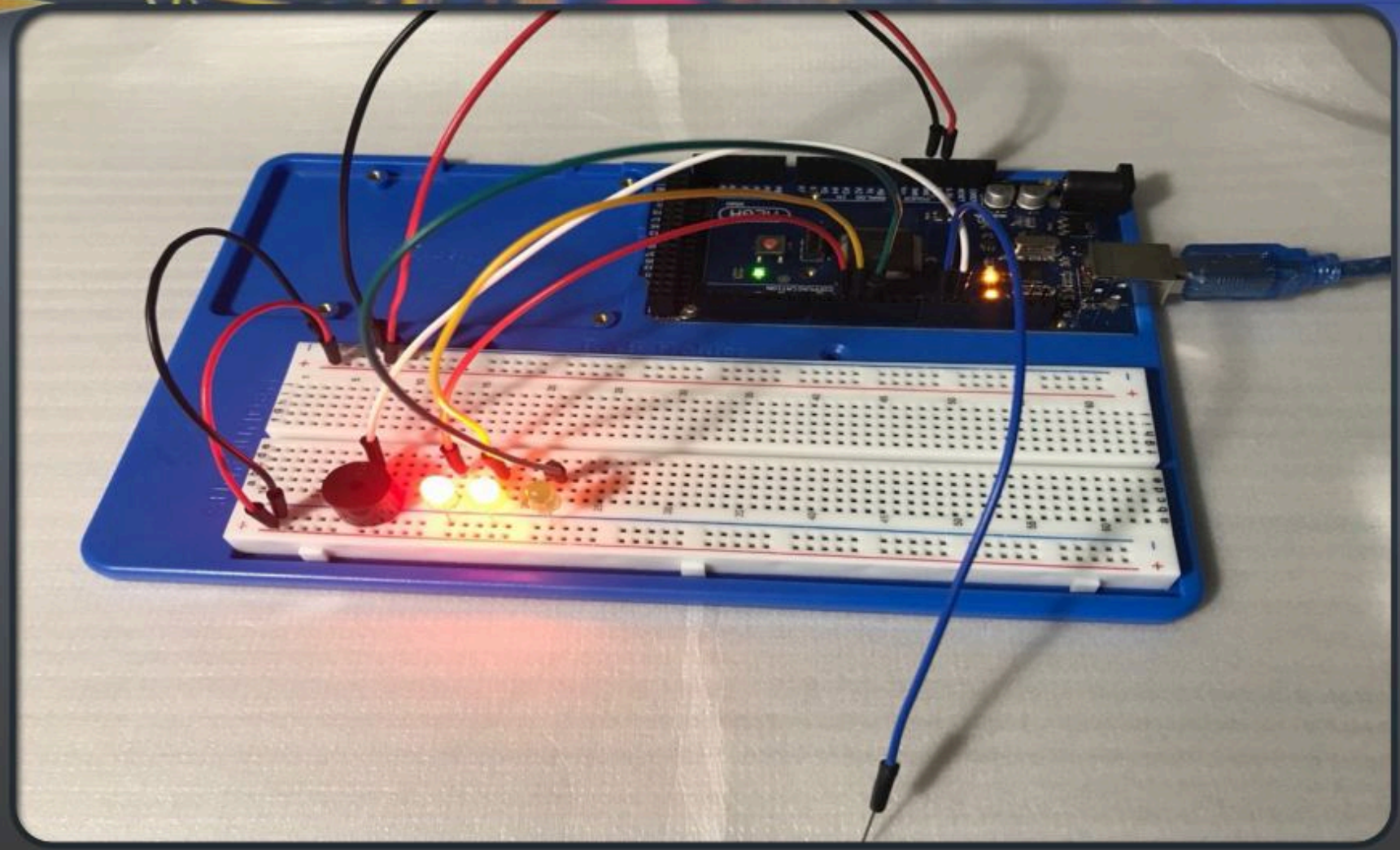
# MODULE 4

## ADDING DOOR SENSOR TO SMART HOME SYSTEM

# CIRCUIT OF DOOR CLOSED WITH GREEN LED ON



# CIRCUIT OF DOOR OPEN WITH GREEN LED OFF



# ARDUINO CODE

CEIS101_4 | Arduino 1.8.19

CEIS101_4 §

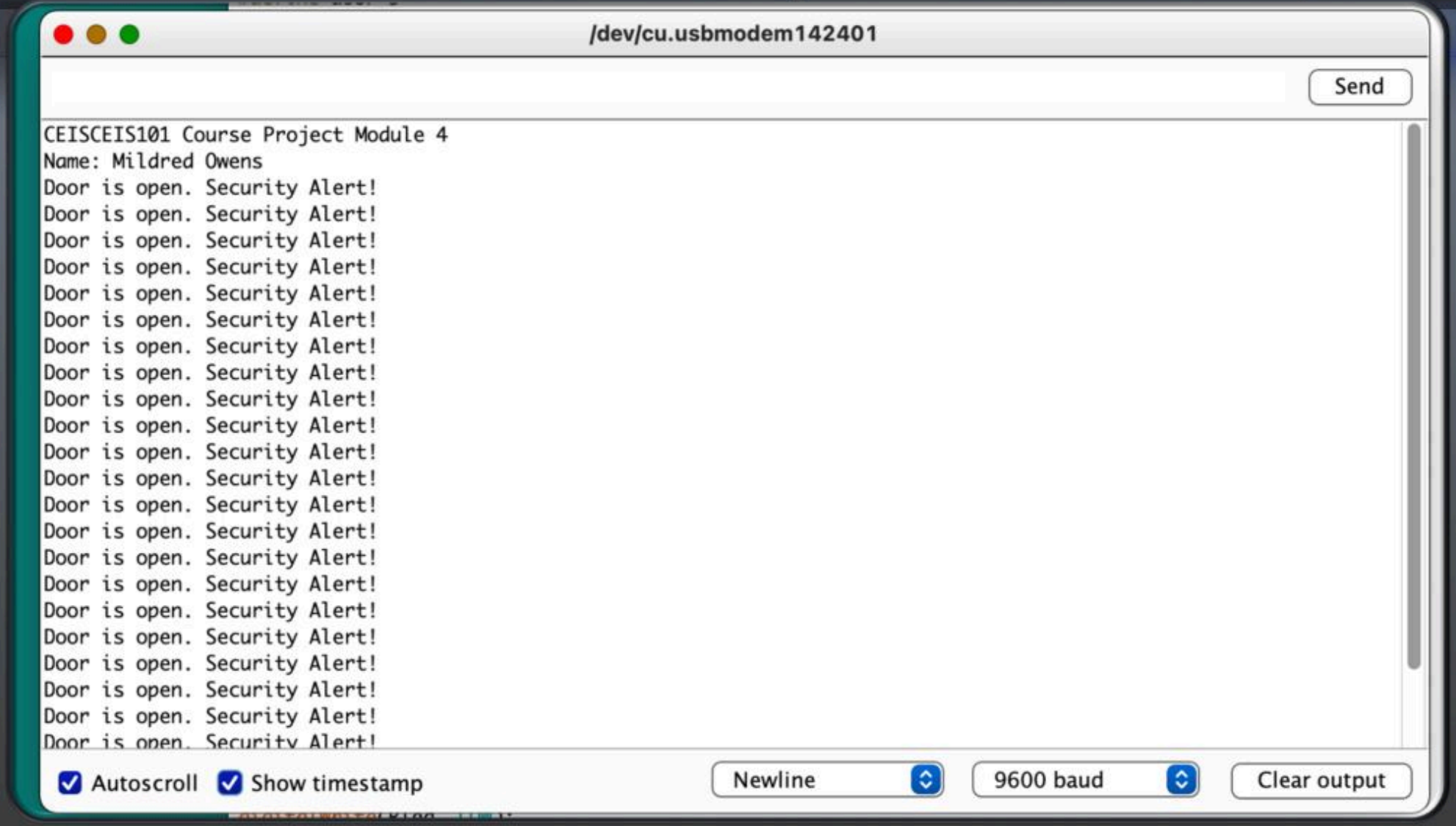
```
#define Rled 2
#define Yled 3
#define Gled 4
#define buzzer 10
#define door 9
#define delaytime 100 // === Second run, change to 100

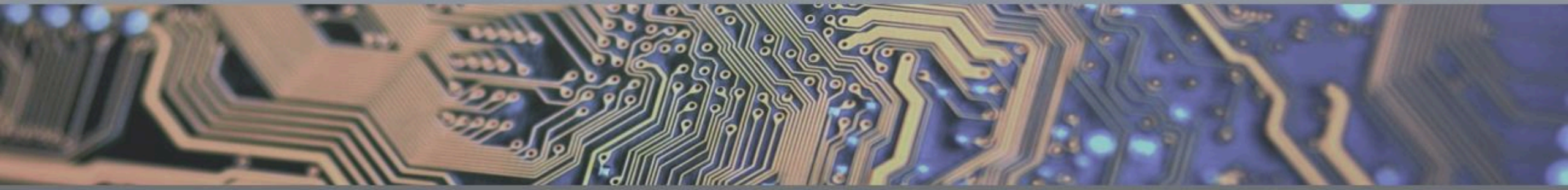
void setup() {
Serial.begin(9600); // Set the baud rate
Serial.println("CEIS101 Course Project Module 4");
Serial.println("Name: Mildred Owens "); //replace xxxxx with your name

pinMode(Rled, OUTPUT);
pinMode(Yled, OUTPUT);
pinMode(Gled, OUTPUT);
pinMode(buzzer, OUTPUT);
digitalWrite(buzzer, LOW);
pinMode(door, INPUT_PULLUP);  //door sensor
}

void loop() {
int value=digitalRead(door);
if(value == 0) {// Door closed, no security threat
digitalWrite(Rled, LOW);
digitalWrite(Yled, LOW);
digitalWrite(Gled, HIGH);
digitalWrite(buzzer, LOW);
}
else{ // Door open, security threat
Serial.println("Door is open. Security Alert! ");
digitalWrite(Rled, HIGH);
digitalWrite(Yled, HIGH);
digitalWrite(buzzer, HIGH);
digitalWrite(Gled, LOW);
delay(delaytime);
digitalWrite(Rled, LOW);
digitalWrite(Yled, LOW);
digitalWrite(buzzer, LOW);
delay(delaytime);
}  // end of else
} //end of loop
```

# SERIAL MONITOR (SCREENSHOT)

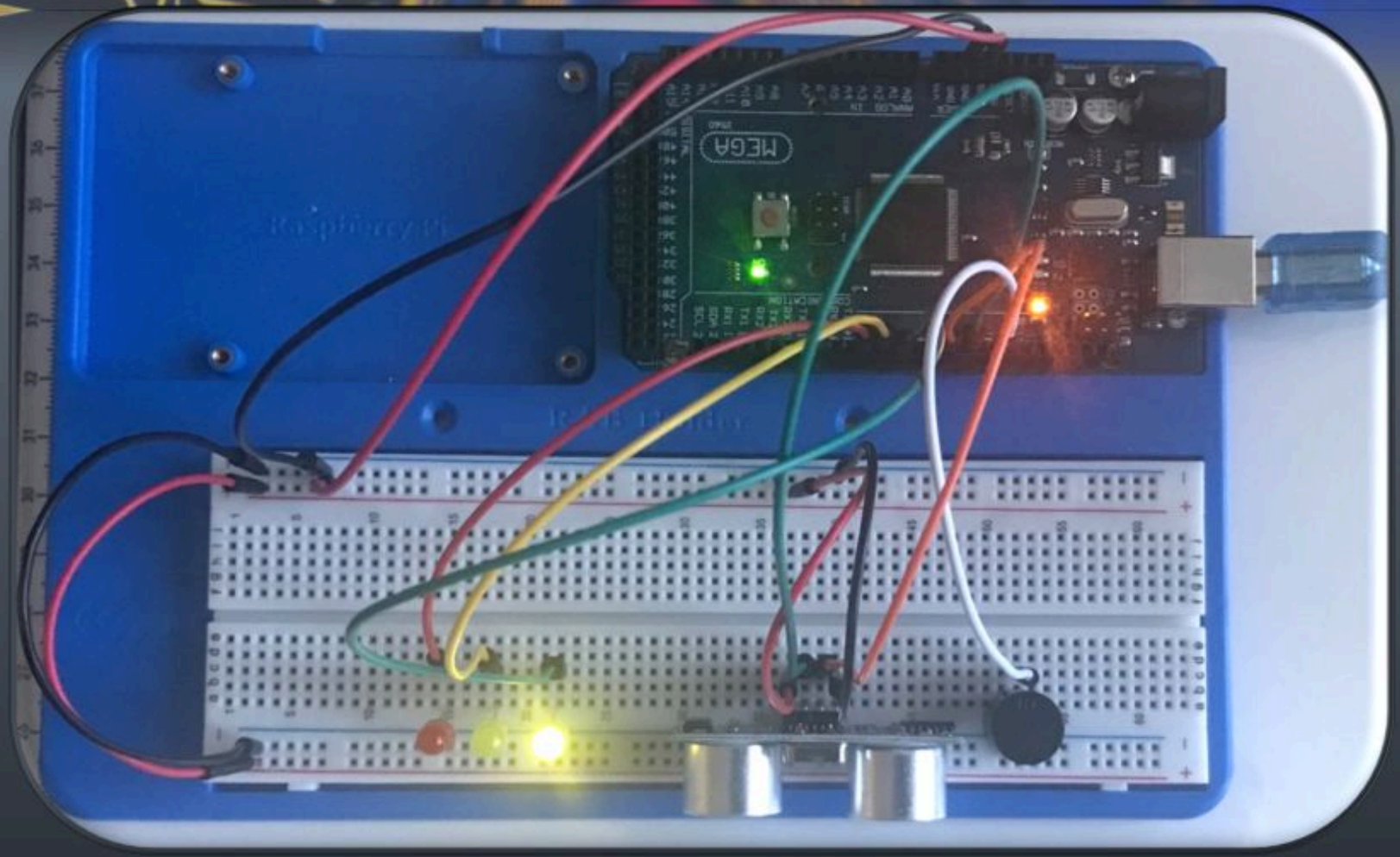




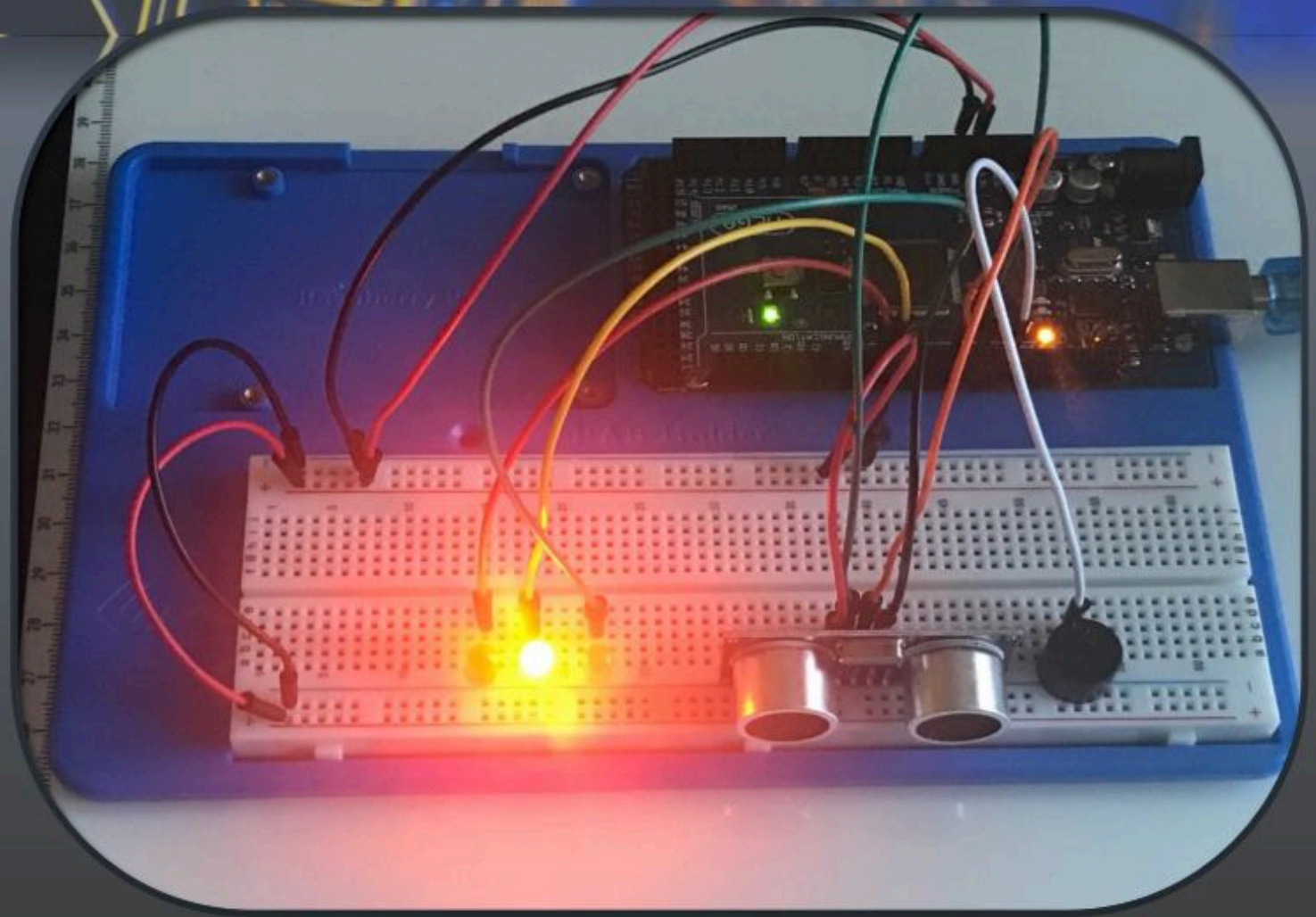
# MODULE 5

## ADDING DISTANCE SENSOR TO SMART HOME SYSTEM AND CONDUCTING DATA ANALYSIS

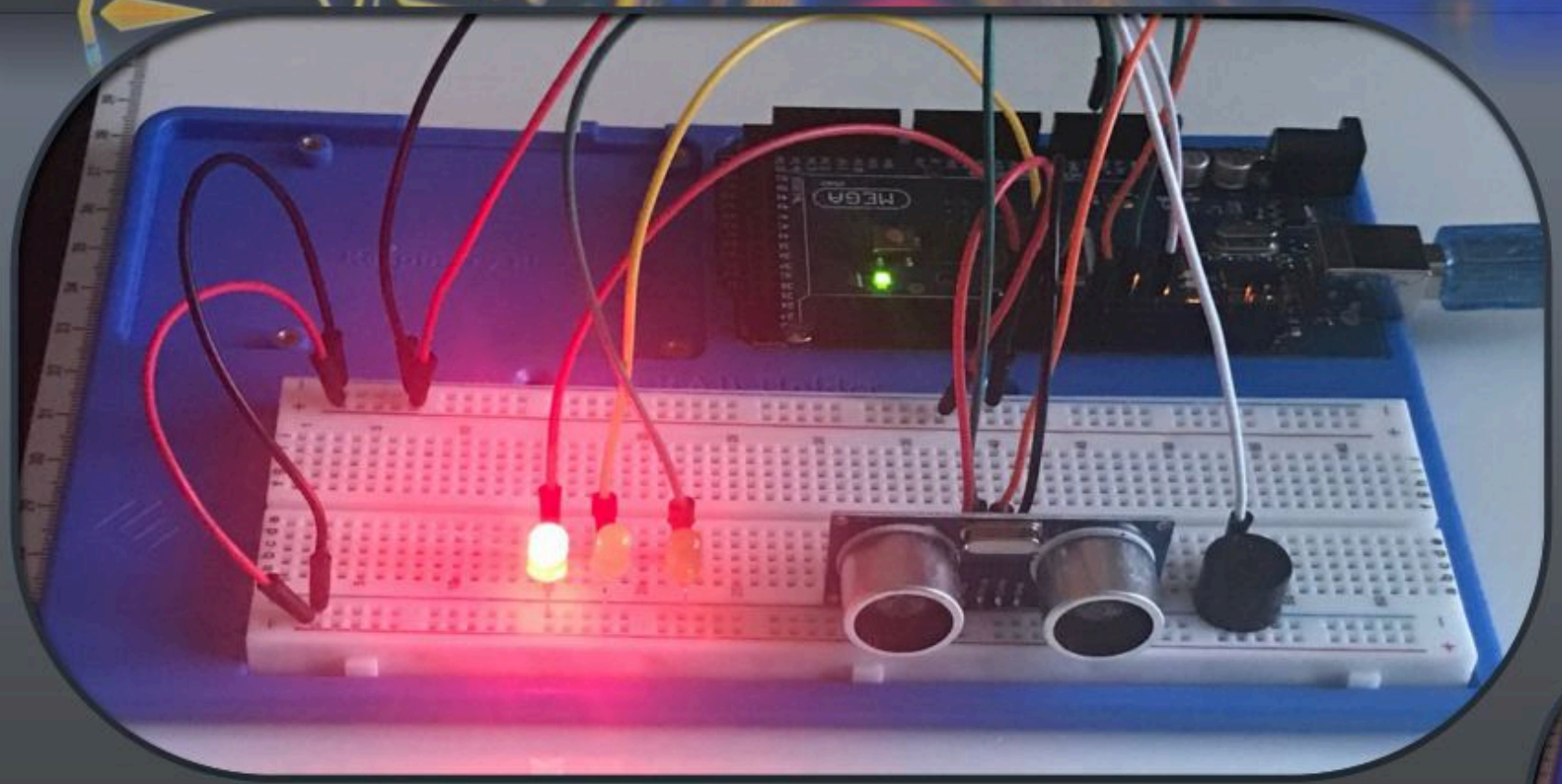
# CIRCUIT WITH GREEN LED ON

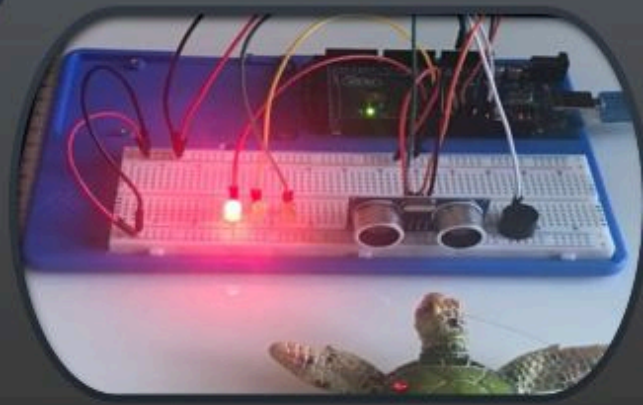


# CIRCUIT WITH YELLOW LED ON



# CIRCUIT WITH RED LED ON



Intruder is at a very close/closest distance based on distance set /programmed for it to set red light on and sound alert (threshold)

# ARDUINO CODE

CEIS101C_5 | Arduino 1.8.19

CEIS101C_5

```
#define trigPin 8
#define echoPin 7
#define Rled 2
#define Yled 3
#define Gled 4
#define buzzer 10

void setup() {
Serial.begin(9600);
Serial.println("CEIS101 Course Project Module 5");
Serial.println("Name: MILDRED OWENS "); //replace xxxxx with your name

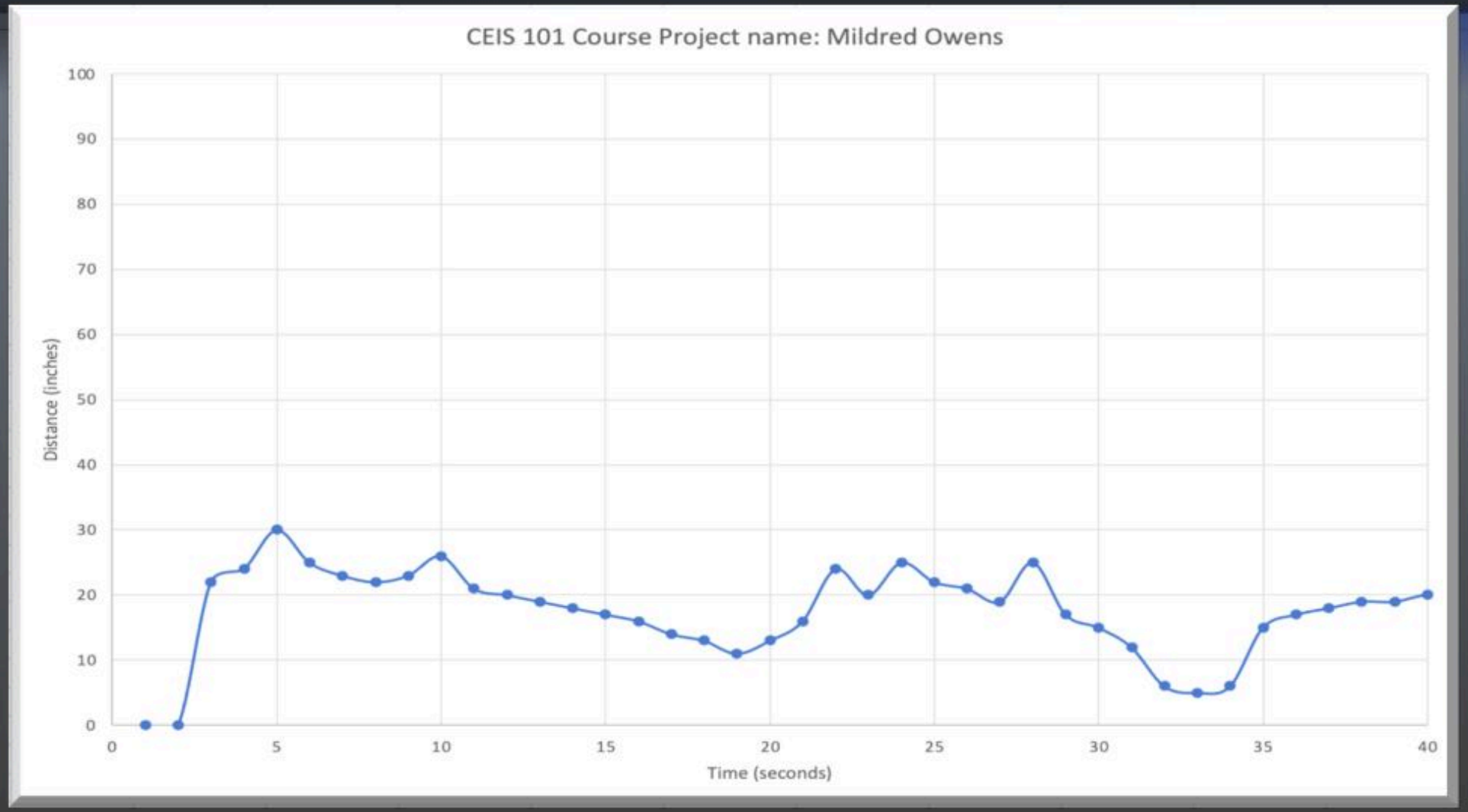
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);
pinMode(Rled, OUTPUT);
pinMode(Yled, OUTPUT);
pinMode(Gled, OUTPUT);
pinMode(buzzer, OUTPUT);
}

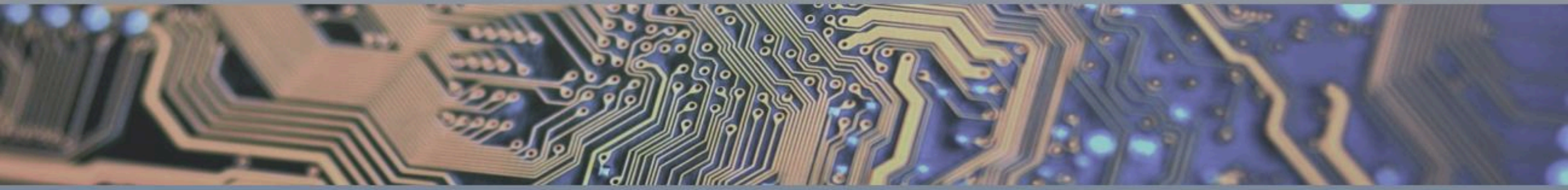
void loop() {
long duration, distance, inches;

digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);

// Read the echo signal
duration = pulseIn(echoPin, HIGH); // Read duration for roundtrip distance
distance = (duration /2) * 0.0135 ; // Convert duration to one way distance in units of inches
```

# PLOT OF DATA (GRAPH FROM EXCEL)

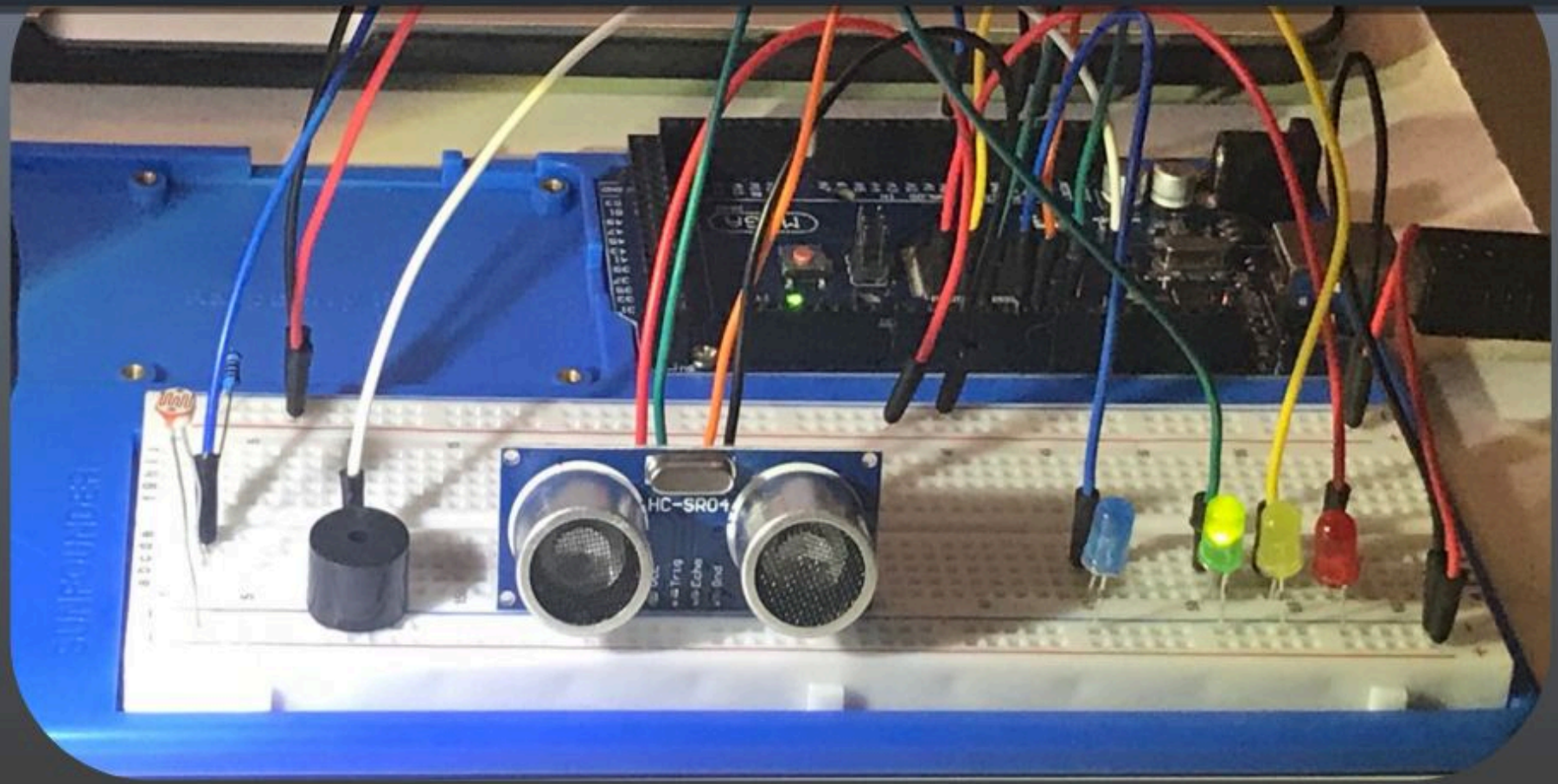




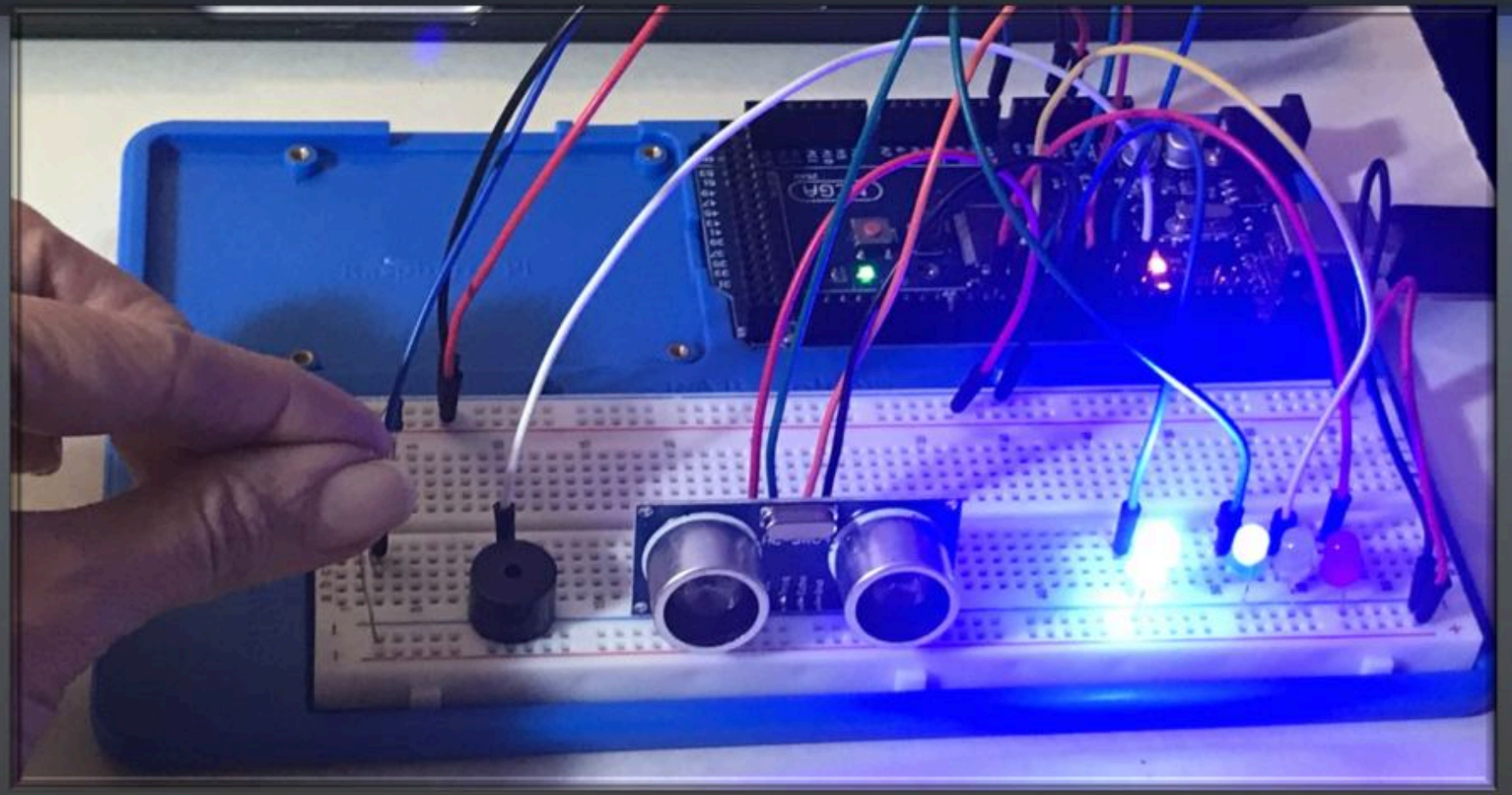
# MODULE 6

## ADDING AUTOMATED LIGHT TO SMART HOME SYSTEM

# CIRCUIT WITH AUTOMATED LED OFF



# CIRCUIT WITH AUTOMATED LED ON



**Photoresistor is a light sensor** often used to detect the presence and absence of light or measure the light intensity like those of night lights. Hovering over = darkness -> will light up.

# ARDUINO CODE

CEIS101_6

```
#define trigPin 8
#define echoPin 7
#define Rled 2
#define Yled 3
#define Gled 4
#define buzzer 10
#define photocell A0
#define autoLight 6

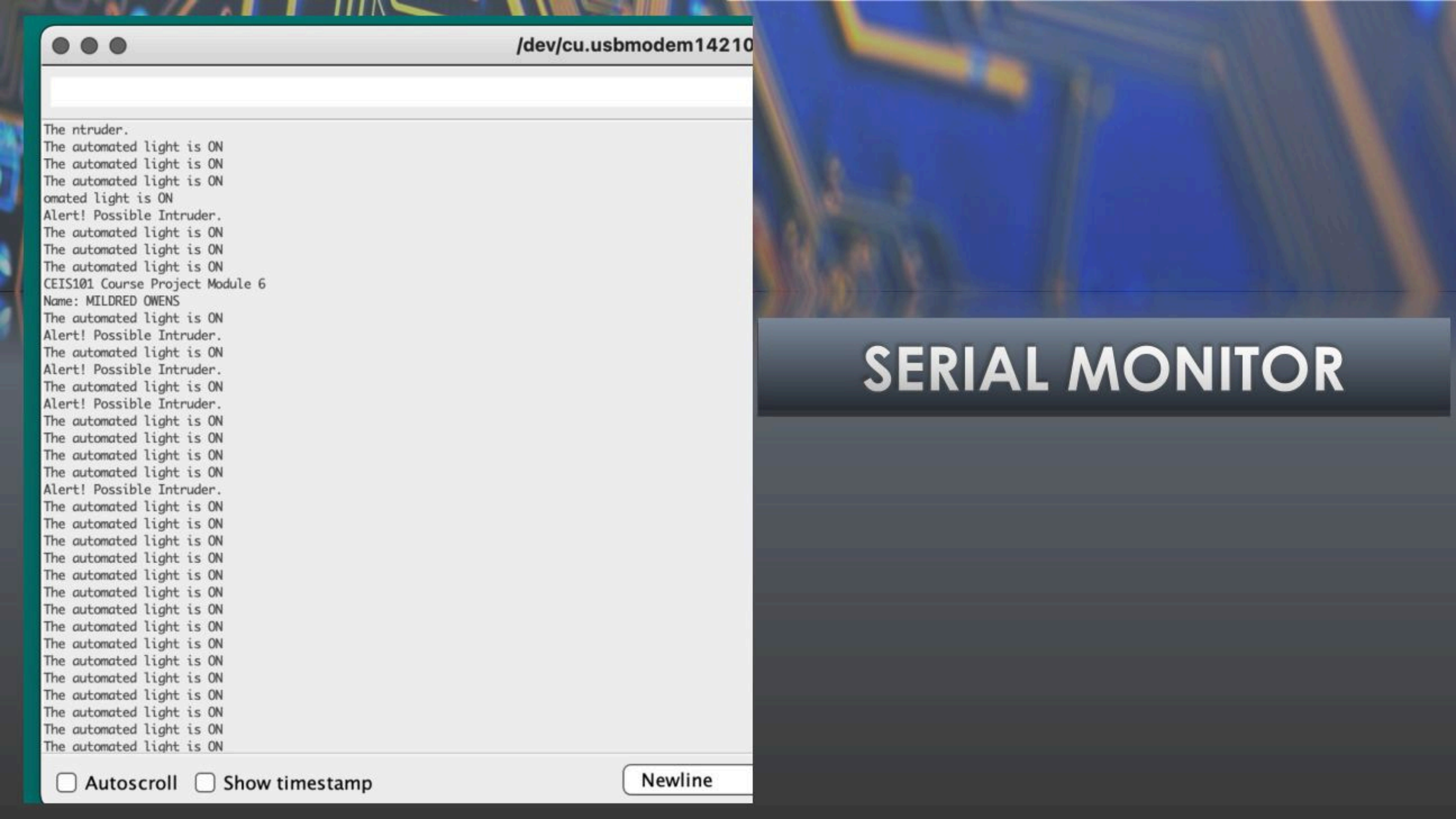
void setup() {
Serial.begin(9600);
Serial.println("CEIS101 Course Project Module 6");
Serial.println("Name: MILDRED OWENS "); //replace xxxxx with your name

pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);
pinMode(Rled, OUTPUT);
pinMode(Yled, OUTPUT);
pinMode(Gled, OUTPUT);
pinMode(buzzer, OUTPUT);
pinMode(autoLight, OUTPUT);
}

void loop() {
//=== Automated Light ===
int value=analogRead(photocell); // Read the value from the light sensor to determine condition

//Serial.println(value); //uncomment this line and open serial plotter to see the effect of light intensit

if (value > 450) {
 digitalWrite(autoLight, HIGH);
 Serial.println("The automated light is ON");
}
else {
digitalWrite(autoLight, LOW);
}
```


/dev/cu.usbmodem14210
The ntruder.
The automated light is ON
The automated light is ON
The automated light is ON
omated light is ON
Alert! Possible Intruder.
The automated light is ON
The automated light is ON
The automated light is ON
CEIS101 Course Project Module 6
Name: MILDRED OWENS
The automated light is ON
Alert! Possible Intruder.
The automated light is ON
Alert! Possible Intruder.
The automated light is ON
Alert! Possible Intruder.
The automated light is ON
The automated light is ON
The automated light is ON
The automated light is ON
Alert! Possible Intruder.
The automated light is ON
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The automated light is ON
The automated light is ON
The automated light is ON
Autoscroll
Show timestamp
Newline
SERIAL MONITOR



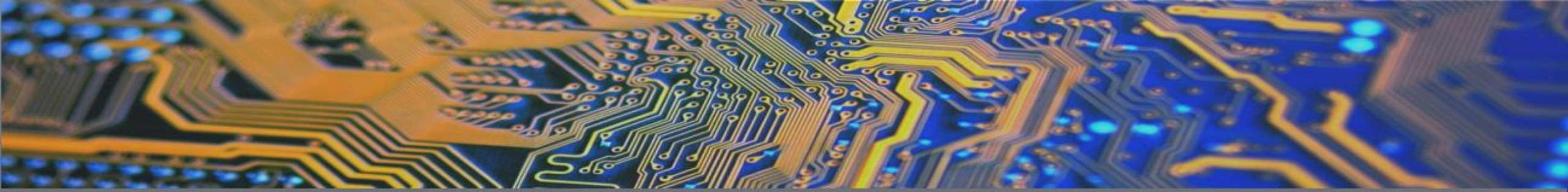
# CHALLENGES/OBSTACLES/SOLUTION

1. Identifying Electronic Parts: I labeled each individual packet and read/studied about the parts description/function and of basic electronics
2. 2 blue led lights didn't light up. I had to learn how to use the tester, since my codes were all correctly loaded. I found out it was faulty/busted.
3. The **USB 2.0B** was too short. I replaced it with a long one to better move around the system.
4. The 1st photoresistor was faulty. I had to troubleshoot wiring, coding, loading, and finally worked with a new photoresister.
5. My wires mostly too short. I ordered medium lengths but received short ones as well. I redesigned my breadboard placement to set up my wirings connections neatly so that they don't get tangled up and connections can easily traceable.
6. Time management: work/job training, housekeeping, dog care, husband needs and occasionally assisted my classmates (facetime).



# Career Skills Gained

1. Developed **problem-solving skills** about the basics of networking, computing, and electronic devices as *applied to IoT* through the building, designing, coding with Arduino Iot Kit automation system project. Hardware+Software+Network and Me
2. Developed understanding of the basics Internet of Things (IoT) and the way that People, Places, Data, and Devices (P2D2) work together.
3. **Networking and Digitization:** Learned the basics of interconnectivity of devices in the IoT. (ARDUINO IDE with the IoT Kit)
4. **Basic Electronics** : Gained some knowledge about Basic Electronics, parts and their functions
5. **Tinkercad** : Learned the purpose of Tinkercad and how to use it.
6. **Arduino IDE**: Basic Insights of C++ for Arduino mega2560
7. **Sharpen skills in Observation, monitoring, and network awareness**



# Conclusion

Entering the field of Tech Core, is indeed challenging and exciting. This course was an adventure for me. As one aspiring to be a computer specialist someday, majoring either in software development and mastering in cybersecurity engineering, I see more clearly how urgent and critical it is to protect information systems and how digitized hardware and network all play together to carry out the goal. The world of Science and Technology evolved leaps and bounds each day. Alongside, more challenges arise for security and further breakthroughs are needed from humans who are the creative minds behind every machine learning and advancement.

All in all, tis  course gave me learning tools and hands on opportunity to have taste on what it could be like designing, building programming a secured home automation system, and appreciate how hardware, software and network all go hand and hand to help us build a better and efficient world of man and machine power.